

System Architecture:

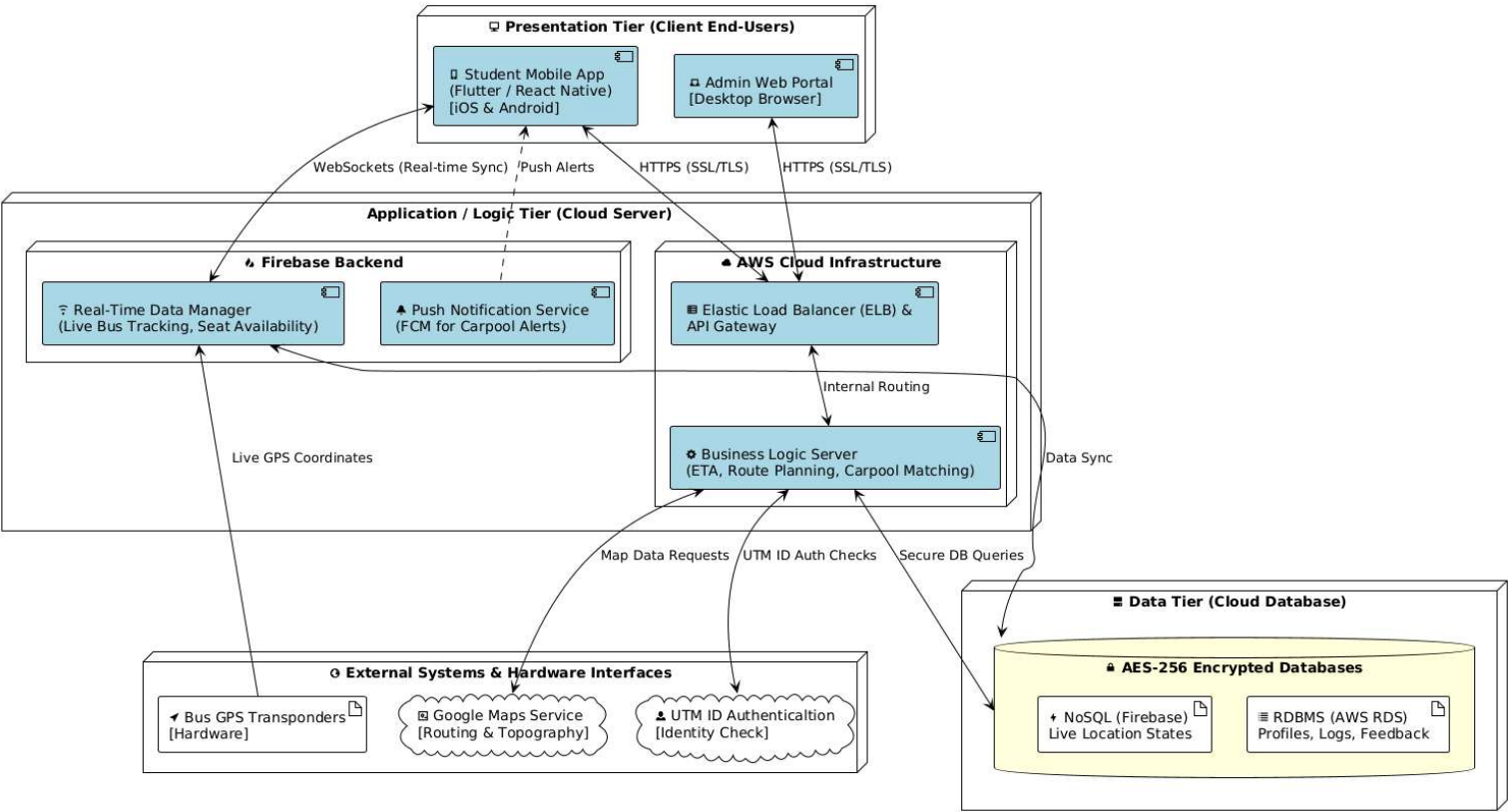


Figure: Overall System Architecture Diagram

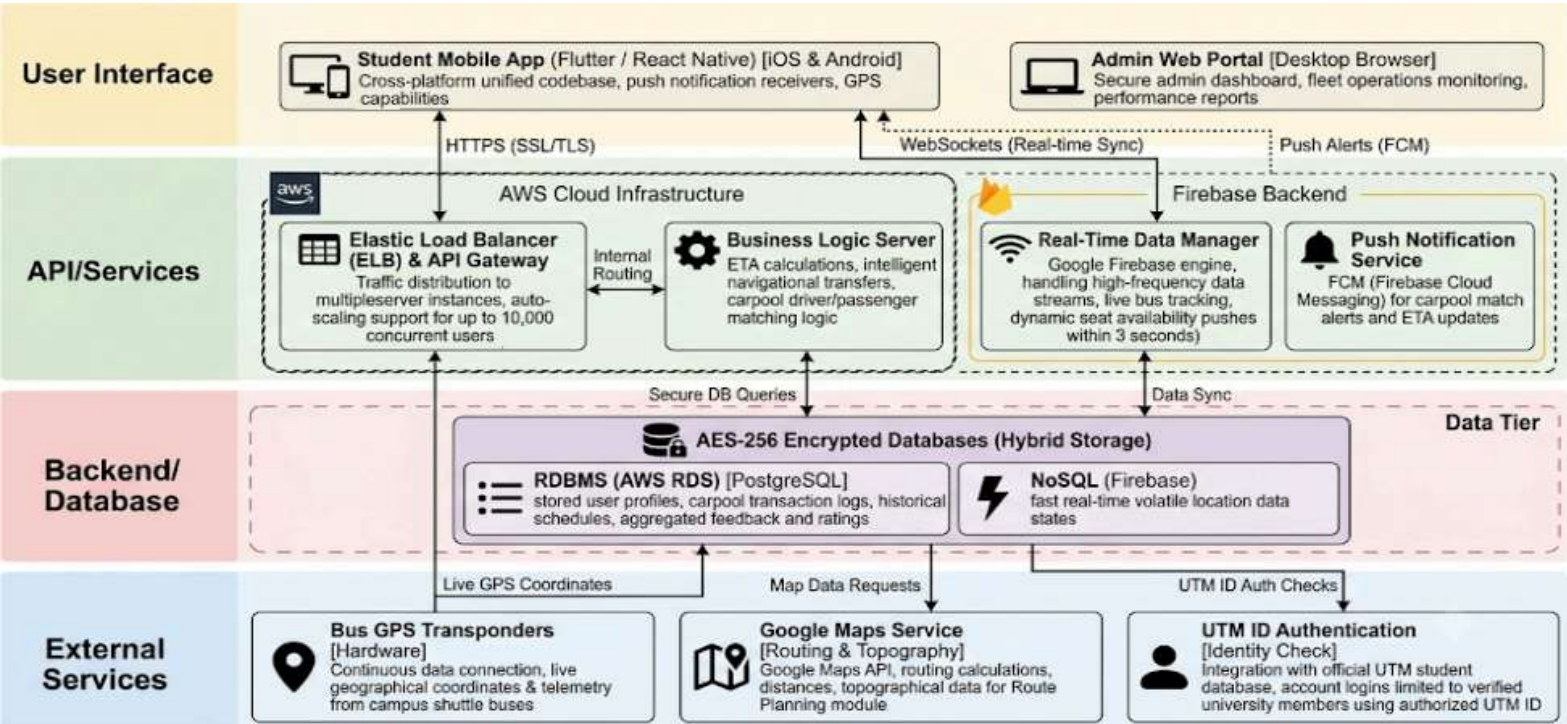


Figure: System Architecture Diagram According to Layer

I. Architecture Overview

UTM Transport App employs a robust, cloud-based, Three-Tier Client-Server Architecture. This structural design guarantees real-time data synchronization, dynamic scalability during peak campus transit hours, and high system availability. By decoupling the presentation, application, and data layers, the architecture ensures maintainability and allows for seamless integration with external hardware and third-party services. All communications between tiers are strictly secured by encryption to protect sensitive data in transit.

II. Presentation Tier (Client/User Interface)

This tier represents the physical end-user environment, ensuring cross-device compatibility and a frictionless user experience.

- **Student Mobile Application(Smartphones):** The primary interface for students utilizes cross-platform software frameworks such as Flutter or React Native. This allows a single, unified codebase to run natively on both iOS and Android operating systems, leveraging the device's built-in internet, push notification receivers, and GPS capabilities without requiring additional hardware purchases by the students.
- **Administrator Web Portal:** The UTM Transport Administrator and IT team interact with the system via a secure, web-based admin dashboard. This interface runs on standard desktop web browsers and allows administrators to monitor network health. Overseeing live fleet operations, and generating monthly performance reports.

III. Application Tier (Server Side)

This tier acts as the brain of the system, hosted entirely on cloud infrastructure to efficiently manage processing and live data distribution.

- **Cloud Hosting and Load Balancing:** Amazon Web Services (AWS) provides the scalable cloud computing infrastructure. AWS allows the system to automatically scale its processing resources. To seamlessly support a huge amount of concurrent users without performance degradation during peak hours, the system utilizes an API Gateway and an Elastic Load Balancer (ELB) to evenly distribute incoming traffic across multiple server instances.

- **Real-Time Data Streams and Notifications:** Google Firebase serves as the real-time backend engine. Firebase is crucial for handling continuous, high-frequency data streams, ensuring that live bus tracking and dynamic seat availability are pushed to student devices within 3 seconds of a change. Additionally, Firebase Cloud Messaging (FCM) is used to send instant push notifications to users when a carpool match is found or an ETA changes.
- **Business Logic Processing:** The server side executes all core computational tasks. This includes calculating precise Estimated Times of Arrival (ETA) based on bus speeds and traffic, generating intelligent navigational transfers, and programmatically matching carpool drivers with passengers based on departure times and destination proximity.

IV. Data Tier (Database)

This tier is responsible for secure, persistent storage of all system records, operating within cloud databases protected by strict data encryption standards to comply with external privacy provisions.

- **Hybrid Data Storage Solution:** The database securely stores and retrieves essential operational entities using a hybrid approach. Firebase's NoSQL database handles the volatile, fast-moving real-time location data, while a Relational Database Management System manages structured, historical data. This includes verified user profiles, a digital log of all carpool transactions, historical bus schedule data, and aggregate student feedback and rating reports. Automated backup protocols are in place to ensure that the probability of data corruption upon failure remains below 1%.

V. External Systems and Hardware Interfaces

To provide full functionality, UTM Transport App integrates with several critical third-party systems and external hardware.

- **Hardware (Bus GPS Transponders):** Physical GPS transponders are installed directly onto all active campus shuttle buses. These hardware devices establish a continuous data connection with the Application Tier, feeding live geographical coordinates and telemetry to accurately plot the fleet on the digital campus map.
- **External API (Google Maps Service):** The system reliably interfaces with the Google Maps API. This third-party integration is utilised by the Route

Planning module to fetch accurate campus topographies, calculate walking and transit distances, and generate step-by-step route calculations.

- **Security and Seamless Authentication (UTM ID Integration):**The system's authentication layer integrates directly with the student's unique UTM ID. This is because this ID is assigned to every student upon enrollment and serves as a universal key for all campus applications, it is already securely logged in the central university database. By linking out application to this existing infrastructure, we maintain a strictly verified, high-trust ecosystem restricted only to active students, while making the login process fast, unified, and effortless.